

Chapter 1

Natural Community Types: Partitioning Natural Variation and Associating with Environmental Descriptors

1.1 Introduction

Before the contamination stress influence on the community can be examined in isolation, the influence of the natural environment must be taken into account. The community records of interest at any point in space and time are the joint outcome of natural environmental conditions, contamination pressure, and other, unrecorded sources of stress(Allan, 2004 [1]). Therefore a site's taxa record is a representation of all these factors acting together.

Two strategies are commonly used: adjusting for the natural environment within the model, or stratifying the data by it before the model is fitted. The first introduces the environmental variables as covariates in the regression of the community records of interest, so that contamination and natural conditions are modelled jointly and estimated finely; its drawback is that contamination gradients typically covary with natural environmental gradients (Graham, 2003 [3]), resulting collinearity with un-

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stable coefficients and limited practical value. The second strategy instead treats the natural environment as a categorical descriptor and partitions the sites into classes of similar environmental and(or) regional—characteristics (Hughes, Larsen, and Omernik, 1986 [4]; Allan, 2004 [1]) to avoid introducing covariates. This stratification can be a more interpretable approach, since it isolates the contamination signal within definable and comparable habitat stratas and divides the community records of interest with natural variation into community types with aligned environmental conditions.

However, the difficulty is practical rather than conceptual. Natural community composition is shaped by a large set of environmental features, many of which are unobservable, observable but not recorded, or recorded only with substantial measurement uncertainty. Consequently, in most empirical settings, the limited set of measured environmental variables is insufficient to define site groups that are genuinely homogeneous in both environmental conditions and the community records of interest. This limitation is illustrated in Figure 1.1. Complete environmental observation aligns the partition in Ω_{env} with the true community boundary; incomplete observation yields only a degraded partition that mislabels the sites between it and the true boundary—an error of incomplete descriptors, not of a failed environment–community relationship.

For this reason, we reverse the order in which environment and community enter the stratification. We first let the community data *define* the types directly, and then ask what environmental signature *describes* them. It rests on the previous conclusion: within the least-contaminated set, the recorded assemblages retain natural community composition. Therefore we can read natural community types from these least-contaminated sites, since they carry natural variation that we wish to stratify by. Starting with the least-contaminated set, we cluster its community matrix, validate and retain K branches as natural community types. Next, we model this environment-assemblage association by training a classifier that maps their environmental descriptors onto the defined community types. Through this process, we uncanonically stratify community types on the taxa space Ω_{taxa} , makes the implicit environment-determination assumption become an explicit and measurable property in the classifier building. This *define-then-describe* workflow is illustrated in Fig-

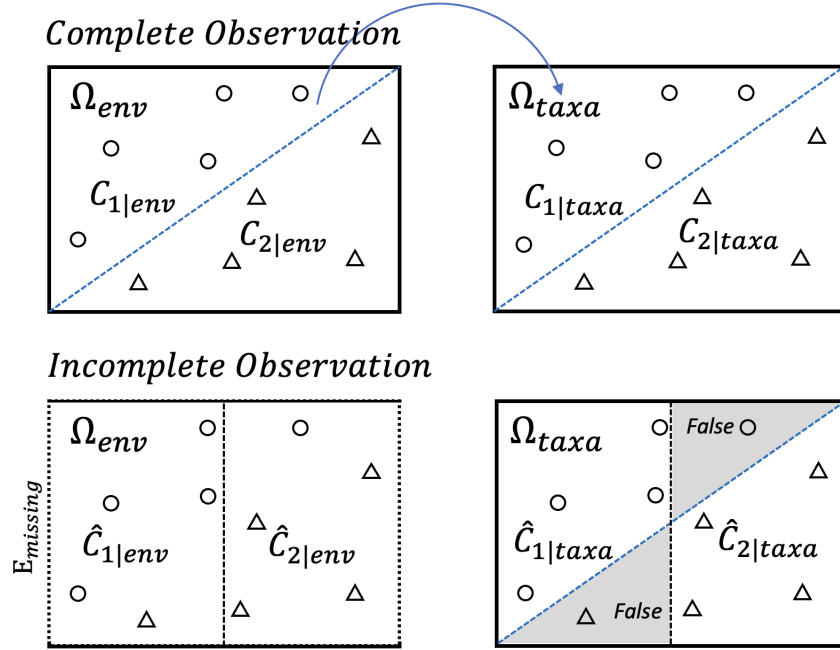


Figure 1.1: Under the assumption that environment determines community, incomplete environmental observation propagates classification error from Ω_{env} to Ω_{taxa} . Complete observation recovers the true diagonal boundary; incomplete observation imposes a degraded vertical partition, mislabelling sites in the wedge between the two (shaded, “False”).

ure 1.2.

The built classifier is then applied to the remaining, contaminated sites: predicting their environmental descriptors into community types. Through this process, every site regardless of contamination status is assigned to its natural community type in a broad sense.

1.2 Methods

The overall aim of this stage is to assign every site, regardless of contamination level, to a natural community type. The procedure has three main steps, the first two carried out entirely within the least-contaminated set and the third applying their results to the remaining sites. First, natural community types are defined on the least-contaminated sites by clustering their community matrix. Second, starting from these

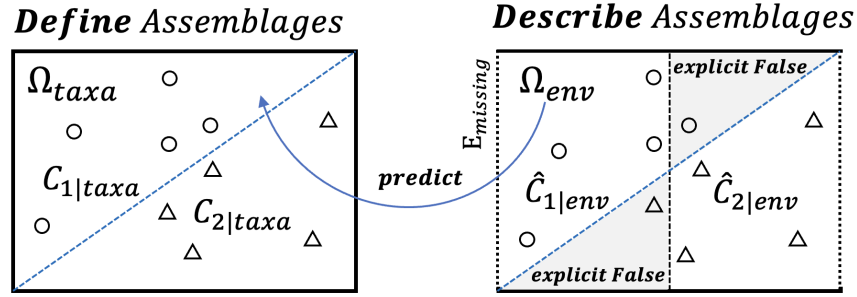


Figure 1.2: Community types are *defined* on the taxa space Ω_{taxa} , then *described* and predicted from the environmental space Ω_{env} . Limited environmental features make some misassignments, but these are rendered explicit (shaded, “explicit False”).

defined types, a discriminating subset of environmental descriptors is identified and a classifier is built to quantify the relationship between environmental features and community types. Third, the classifier is applied to the remaining contaminated sites, predicting and delineating the decision boundary by which they are assigned to the corresponding natural community type. The remainder of this section describes each step in detail.

1.2.1 Unconstrained community typing and its statistical validation

Dissimilarity between sites, Double-zero problem and Hellinger transformation

Hierarchical clustering operates on a matrix of pairwise dissimilarities, so the first choice is how dissimilarity between sites is measured from the community records. Applying the Euclidean distance directly to raw abundances is inappropriate for community data, because it treats the joint absence of a taxon at two sites as evidence of similarity—the *double-zero problem*—and lets a few abundant taxa dominate the distance. To avoid this, the community matrix is first Hellinger-transformed: at a given site, each taxon’s record is divided by the site’s total and the square root is taken, so that the entry becomes the square root of the taxon’s relative abundance at that site. The Euclidean distance computed between two Hellinger-transformed site

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profiles is then equal to the Hellinger distance between their original compositions, a measure that is insensitive to double zeros and to differences in total abundance, and which is recommended for ordination and clustering of community data in numerical ecology (Borcard, Gillet, and Legendre, 2018 [2]). This transform therefore makes the subsequent Euclidean-based clustering ecologically meaningful.

Hierarchical clustering with Ward’s method

The Hellinger-based dissimilarities are clustered by agglomerative hierarchical clustering. Among the available linkage functions, Ward’s method is chosen because it merges, at each step, the pair of clusters whose fusion produces the smallest increase in within-cluster variance of the Euclidean (here Hellinger) distances. This criterion suits the goal of delineating community types with members mutually similar in composition. The procedure yields a dendrogram in which the height of each fusion records the linkage function value across which the corresponding branches join; cutting the dendrogram at a chosen linkage height then defines the K community types retained for the subsequent analysis.

Statistical evidence for the branching: node-wise and overall ANOVA

A dendrogram will partition any data, so the branching structure requires statistical support. Two complementary tests are used, both applied to the Hellinger-transformed taxa abundances used in the clustering rather than to the raw counts. First, a *node-wise* ANOVA is applied from the first to $K + 2$ branching nodes of the dendrogram: for every taxon, its Hellinger-transformed abundances in the two daughter branches are compared, identifying the taxa whose mean values differ significantly across the split. These taxa are the indicator species that contribute most to the formation of that particular node and drive the bifurcation. Second, once the K community types are finalised, an overall ANOVA tests, for each taxon, whether its Hellinger-transformed mean abundance differs significantly among the K groups, providing statistical evidence that the retained partition separates the community types along interpretable compositional gradients.

Robustness of the partition with pvclust

Beyond the compositional evidence for each branch, the stability of the branches under resampling is assessed with pvclust (Suzuki and Shimodaira, 2006 [5]; Suzuki, Terada, and Shimodaira, 2019 [6]). The method repeatedly perturbs the descriptor space by multiscale bootstrap resampling of the taxa, re-runs the same clustering procedure on each resampled space, and checks whether each branch of the original (anchor) dendrogram reappears. From the frequency with which a branch recurs across the B bootstrap replicates, an *Approximately Unbiased* (AU) p -value is computed for that branch. A high AU p -value means the branch is recovered in a large proportion of the perturbed spaces—its existence does not depend on a few particular taxa but is supported by many. The ordinary *Bootstrap Probability* (BP), estimated from fixed-size bootstrap resampling alone, is also reported, but it is a more biased than the AU value. Branches with low AU support are treated as uncertain and inform the choice of where the dendrogram is cut. Figure 1.3 illustrates the computation workflow¹, more details about this procedure can be found in the articles by Suzuki and Shimodaira (Suzuki and Shimodaira, 2006 [5]).

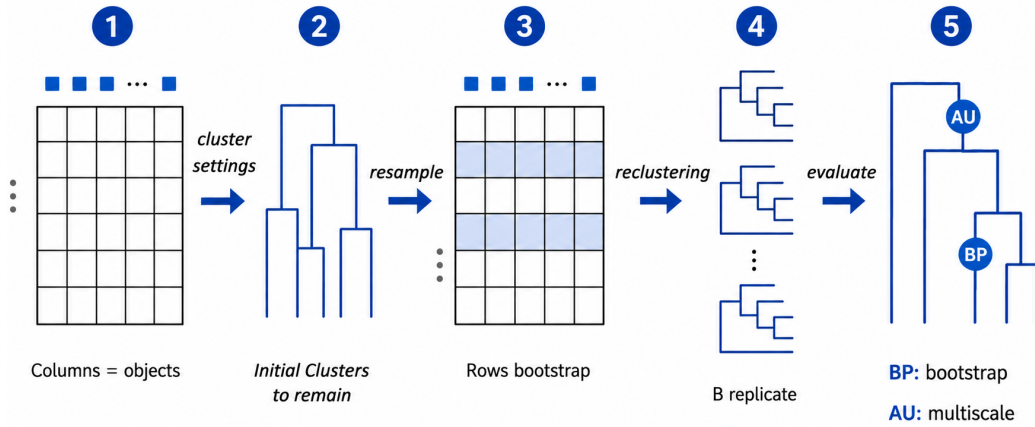


Figure 1.3: The pvclust workflow: bootstrap-resample taxa, recluster B times, and score each anchor branch by its AU (multiscale) and BP (bootstrap) recurrence.

¹The data matrix is transposed so that taxa occupy the rows, because resampling routines in R and similar environments operate row-wise; bootstrapping the feature variables therefore requires placing them as rows.

1.2.2 Linking community types to environmental descriptors: classifier training under limited features

Limited environmental features and the 2008 reference-typing solution

A key difficulty persists as seen in environmental stratifying strategy still persists in this design: the limited set of measurable environmental features. Types delineated by unconstrained clustering may remain entangled in environmental space, where one or more missing key variables, or attention to uninformative ones, blurs the environment–taxa cluster relationship. This already surfaced in the 2008 corridor study (Zhang, 2008 [7]).

There, the Ward-clustered community types could not be cleanly separated on the environmental descriptors, so a directly trained classifier would have performed poorly. The solution was to define the two types by a selective assignment of branches. As shown in Figure 1.7, the dendrogram first divides into B_2 versus the pair $\{B_1, A\}$, and at a lower height do B_1 and A separate. Cluster C_1 was assigned to B_2 and B_1 jointly, and Cluster C_2 to A alone—a hindsight operation under which the types separate more readily on the available descriptors.

The cost, however, is that the types no longer reflect the dendrogram. The assignment is *non-monophyletic*: C_1 joins B_2 and B_1 , two branches that do not form a single sub-tree, while the topologically closer A is placed in C_2 . No single horizontal cut can produce this grouping; the labels are assigned across the tree’s natural splitting order. This solution thus lets environmental separability override the community structure the clustering reveals (essentially induced the environmental-stratifying approach). As recognized an explicit property to reflect the environment-determination assumption, the classifier’s resulting accuracy is inflated and loses its realistic value. (Figure 1.8).

In the environmental descriptor-space view, this operation amounts to relabelling the clustering-defined sites: circles (C_1) that sit, on the available features, among the triangles (C_2) are reassigned to C_2 , buying environmental separability at the cost of the true taxa labels (Figure 1.4 ²).

²The true boundary (blue dashed) is governed by unobserved variables $E_{missing}$ so it is not a diagonal but leans to the $E_{available}$ direction.

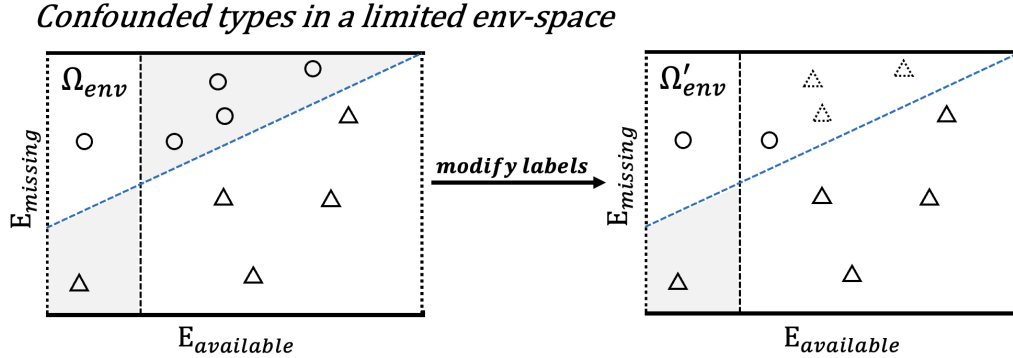


Figure 1.4: Confounded community types in a limited environmental space, illustrating the 2008 approach of Zhang. This solution relabels part of C_1 (circles) as C_2 (triangles) in one direction only ($\Omega_{env} \rightarrow \Omega'_{env}$, dotted triangles), reducing but not removing the overlap—separability bought by altering the true cluster labels.

Classification complexity and the environmental descriptor subset

The 2008 approach is therefore mixing the

because it resolved the limited-feature problem on the wrong side of the relationship: it adjusted the community labels to suit the environment. Yet the underlying obstacle—that the taxa-defined types may not separate on all the measured environmental variables—remains, and must be confronted without touching the type definitions. The question, then, is what to do on the environmental side. The answer is not to feed the classifier every available descriptor, but to recognise that only some descriptors actually discriminate the defined types while others merely add noise, and to select the effective subset accordingly.

A general relationship governs this step from defining community types to selecting environmental descriptors: the more finely the types are defined, the more environmental features are needed to separate them, and a coarser typology can be resolved by fewer features. Although the type labels follow from the clustering algorithm, the number of types retained—where the dendrogram is cut—is ultimately a subjective choice of the analyst, so the descriptor subset required depends on how complex a typology we choose to impose. Rather than altering the type definitions to fit the environment, we leave the taxa-defined types untouched and instead search the environmental side for the descriptor subset that best reflects them. We modify

the descriptor space, never the community labels.

The motivation is that not all environmental features help separate a given typology, and some only add noise. Consider separating aircraft from cars by spatial position. Longitude and latitude can be deceptive: an aircraft over a city, or a car near an airport, share the same horizontal coordinates, whereas the same aircraft over the ocean would appear well separated by them. The discriminating power of these features therefore depends on the sampling distribution—a sample mixing both situations leaves longitude and latitude only weakly discriminating. Removing them and keeping the single key feature, elevation, separates the two classes cleanly: fewer but better-chosen descriptors give stronger separation (Figure 1.5).

In our setting, the community types are fixed first, and a feature-selection step then seeks a compact, effective environmental space in which these fixed types separate as well as the available features allow. Within the selected space, each site is assessed for the consistency between its taxa-defined type and its environmental position: sites whose environmental neighbourhood agrees with their type are *concordant*, while the minority whose environmental position conflicts with their type are flagged as *conflicting* (Figure 1.6). Crucially, conflicting sites are not relabelled to improve separability, as in the 2008 approach; they are retained, as the direct expression of the remaining limitation of the available environmental features.

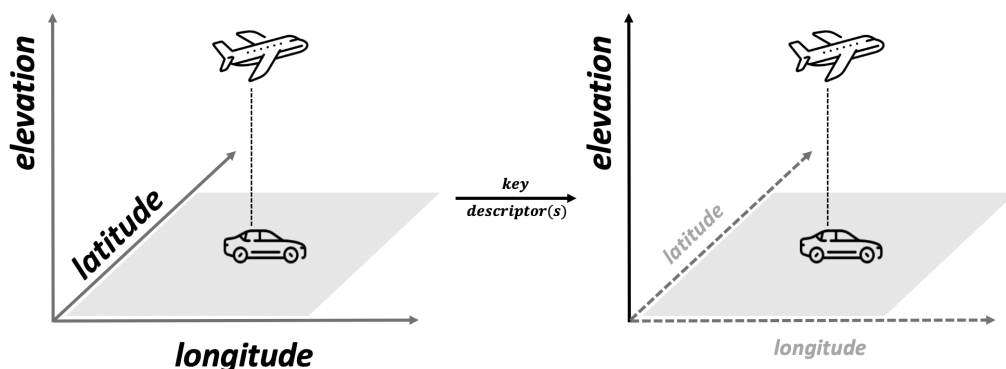


Figure 1.5: Not all descriptors help separate a small number of classes. An aircraft and a car are barely distinguishable on the horizontal descriptors (longitude, latitude), yet a single key descriptor—elevation—separates them cleanly. When only a few classes must be told apart, the right descriptor subset matters more than the number of descriptors, and retaining irrelevant ones only adds noise.

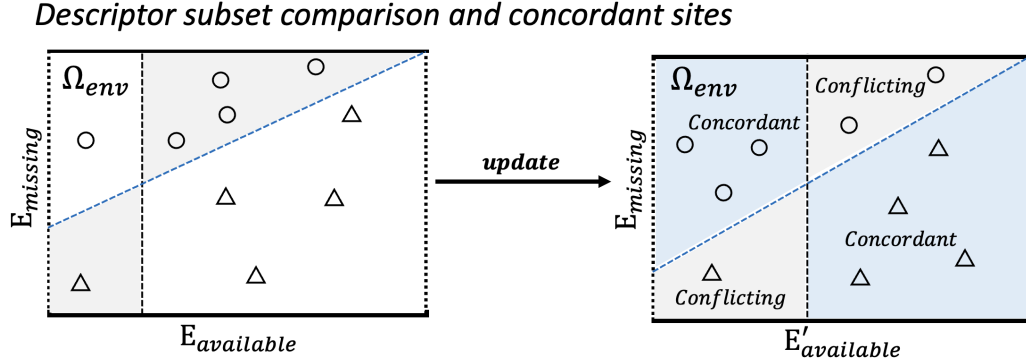


Figure 1.6: Selecting a descriptor subset and identifying concordant sites. On the initial descriptors $E_{available}$, the taxa-defined types overlap (left). Choosing a more discriminating subset $E'_{available}$ (right) separates most sites into *concordant* regions (blue), where taxa membership and environmental position agree, while the residual *conflicting* regions (grey) mark sites whose two views disagree. Only concordant sites are used to train the classifier; conflicting sites are retained as diagnostics. In practice, concordance is resolved across four categories (concordant, taxa-only, env-only, ambiguous), here simplified to two.

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REFERENCES

1.2.3 Appendix

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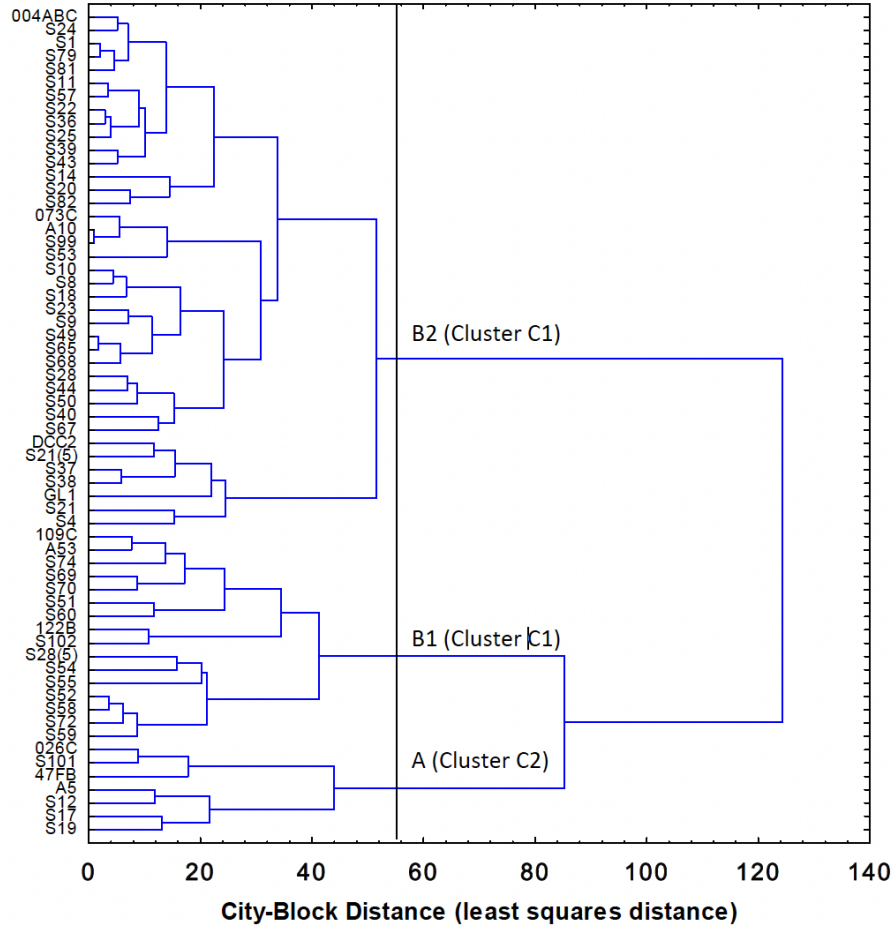


Figure 2 Dendrogram of REF sites ($n = 62$) grouped according to similar zoobenthic community composition in the 1991, 1999 and 2004/5 Lake Huron-Lake Erie Corridor analysis (Ward's method clustering city-block distances of octave-transformed relative abundances of zoobenthic taxa).

Figure 1.7: Branch-selective definition of community types in the 2008 reference dendrogram. The tree first separates B_2 from $\{B_1, A\}$, and only at a lower height splits B_1 from A . Cluster C_1 is assigned to B_2 and B_1 jointly, and Cluster C_2 to A alone—a non-monophyletic grouping that no single horizontal cut can produce, since A is topologically closer to B_1 than B_2 is.

REFERENCES

Table 2.5. Summary of observed number of Lake Huron-Lake Erie Corridor sites in each cluster (columns) identified by zoobenthic taxa relative abundances and membership predicted (rows) by discriminant function classification (Appendix III) on the basis of habitat characteristics measured at those sites

Group	% Correct	Observed	
		Cluster C1	Cluster C2
Cluster C1	98	54	1
Cluster C2	71	2	5
Total	95	56	6

Figure 1.8: The confusion matrix of a LDA classifier trained in the least-contaminated sites with their environmental descriptors and the 2008 branch-selective types. The unbalanced distribution of types (more C_1 than C_2) and extremely high accuracy is caused by subjectively assigning labels in a hindsight manner to achieve environmental separability.